THE CHINESE REMAINDER DECOMPOSITION AND HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Henselization of a pair (A, I) and its universal property. defined and discussed in the section, source lines 3180-3250, immediately before the counted statement. Supplies the object whose behaviour under a product decomposition is asserted.
- Chinese remainder theorem for a ring and two ideals generating the unit ideal. used at several points of the proof, notably around proof lines 3290-3310. Provides the product decomposition of the rings involved.
- The characterisation of a henselian pair through the lifting of factorisations of monic polynomials. used in the proof to verify the henselian property of the decomposed rings. Gives the property that has to be checked for each factor.

Source reference. Original lemma 0H7Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21465–21465. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0A02: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12620–12620. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04E7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5753–5753. Navigation pointer to a statement the

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BBY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14261–14261. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00DT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 509–509. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 04E6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5752–5752. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $I_1, I_2, \ldots, I_n \subset A$ be ideals in a ring A. Suppose $I_i + I_j = A$ for $i \neq j$, $1 \leq i, j \leq n$. Denote A^h the henselization of the pair $(A, I_1 \cap \ldots \cap I_n)$ and A_i^h the henselization of the pair (A, I_i). Then the natural morphism*

$$A^h \to \prod\nolimits_{i=1,\ldots,n} A_i^h$$

is an isomorphism.

Proof. The map comes from the functoriality of the construction. If $n = 1$ the result is immediate. If $n > 1$, observe that $(I_1 \cap \ldots \cap I_{n-1}) + I_n = A$ by our condition on the ideals. Thus if we prove the lemma for $n = 2$, then the lemma follows for all n by induction.

The case $n = 2$. Choose $f_i \in I_i$ such that $f_1 + f_2 = 1$. Recall that A^h is constructed as the colimit of A-algebras B such that $A \to B$ is étale and $A/I_1 \cap I_2 \to B/(I_1 \cap I_2)B$ is an isomorphism. See proof of Lemma 0A02. Denote $Hens$ the category of such A-algebras B. Similarly, denote $Hens_i$, $i = 1, 2$ the category of étale A-algebras B such that $A/I_i \to B/I_iB$ is an isomorphism.

Consider the subcategory $Hens_1(f_2) \subset Hens_1$ consisting of those B such that f_2 is invertible. This is a cofinal subcategory since if B is in $Hens_1$, then B_{f_2} is in $Hens_1$ as well (hint: $B_{f_2}/I_1B_{f_2} = (B/I_1B)_{f_2} = B/I_1B$ because f_2 is a unit modulo I_1). Similarly, the subcategory $Hens_2(f_1) \subset Hens_2$ is cofinal. Hence to compute A_1^h and A_2^h we may work with this smaller categories, see Categories, Lemma 04E7.

Consider the functor

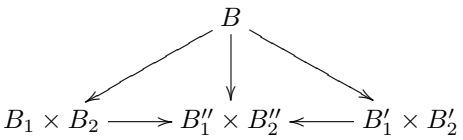
$$Hens_1(f_2) \times Hens_2(f_1) \to Hens, \quad (B_1, B_2) \mapsto B_1 \times B_2$$

This functor makes sense as

$$\begin{aligned}(B_1 \times B_2)/(I_1 \cap I_2)(B_1 \times B_2) &= B_1/(I_1 \cap I_2)B_1 \times B_2/(I_1 \cap I_2)B_2 \\ &= B_1/I_1B_1 \times B_2/I_2B_2 \\ &= A/I_1 \times A/I_2 \\ &= A/(I_1 \cap I_2)\end{aligned}$$

The second equality holds because $(I_1 \cap I_2)B_1 = I_1B_1 \cap I_2B_1 = I_1B_1$ as $f_2 \in I_2$ is invertible in B_1 (we have also used Algebra, Lemma 0BBY). The last equality holds by Algebra, Lemma 00DT. To finish the proof, it suffices to prove that the displayed functor is cofinal (see lemma quoted above). The first condition of Categories, Definition 04E6 holds because an object B of $Hens$ maps to $B_{f_2} \times B_{f_1}$

which is in the image of the functor. The second one always holds because if we have $B \to B_1 \times B_2$ and $B \to B_1' \times B_2'$ with $B_1, B_1' \in Hens_1(f_2)$ and $B_2, B_2' \in Hens_2(f_1)$, then we can set $B_1'' = B_1 \otimes_B B_1'$ which is in $Hens_1(f_2)$ and similarly $B_2'' = B_2 \otimes_B B_2''$ in $Hens_2(f_1)$ to find a map $B \to B_1'' \times B_2''$ fitting into the commutative diagram



This finishes the proof. □